

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	47.0 / Female
Department	General Surgery
Diagnosis	Bladder infection
UTI Classification	Complicated UTI
Sample Type	Suprapubic Aspirate

Clinical Presentation

Chief Complaints: Lower abdominal pain;Dysuria

Comorbidities: Diabetes

Surgical History: Hysterectomy

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	33.0	%	20 - 40
Wbc	9500.0	cells/ μ L	4000 - 11000
Polymorphs	63.0	%	40 - 75
Crp	11.0	mg/L	< 10
Rft Serum Creatinine	1.0	mg/dL	0.6 - 1.3
Serum Uric Acid	4.4	mg/dL	3.5 - 7.2
Blood Urea	27.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	4.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	2.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Candida species
Bacteria Type	Fungus (Yeast; general category for non-albicans or unidentified)

Antibiotic Susceptibility Profile

Predicted Sensitive	Fluconazole, Voriconazole
Predicted Resistant	Ciprofloxacin

Recommended Therapy

Fluconazole

Dosage: 200 mg orally once daily for 7 days

Precautions: Renal function normal; no dose adjustment needed. Monitor liver enzymes periodically. Avoid use in pregnancy or if patient has known hypersensitivity to azoles. Check for drug interactions, especially with drugs metabolized by CYP2C19.

Rationale: Fluconazole is the first-line antifungal for uncomplicated and complicated bladder Candida infections, with good urinary tract penetration and a favorable safety profile.

Voriconazole

Dosage: 200 mg orally twice daily for 7 days

Precautions: Monitor liver function tests and visual acuity; avoid in severe hepatic impairment. Warn of photosensitivity and possible visual disturbances. Check for interactions with CYP2C19 inhibitors or inducers. No dose adjustment for mild renal impairment.

Rationale: Voriconazole is an alternative for Candida infections when fluconazole resistance is suspected or when a broader spectrum is required; it provides excellent activity against many non-albicans species.

Antibiotic Drug History

Clinical Summary

Patient: 47-year-old diabetic female, post-hysterectomy, presenting with lower abdominal pain and dysuria.
Diagnosis: complicated UTI (interstitial cystitis) caused by a bladder infection. Predicted pathogen: Candida species (yeast). Resistance profile: ciprofloxacin resistant. Sensitive agents: fluconazole and voriconazole.

Recommended regimen

- **Fluconazole** 200 mg PO once daily for 7 days - first-line for Candida UTIs, excellent urinary tract penetration, no dose adjustment in normal renal function. Monitor liver enzymes, avoid pregnancy, check CYP2C19 interactions.
- **Voriconazole** 200 mg PO twice daily for 7 days - alternative when fluconazole resistance is suspected or a broader spectrum is needed (effective against non-albicans Candida). Monitor LFTs, visual acuity, photosensitivity; avoid severe hepatic impairment; watch CYP2C19 interactions.

Key precautions: ensure hepatic function, avoid pregnancy, monitor drug interactions, consider diabetes and renal status. Rationale: both agents have proven urinary tract penetration and efficacy; fluconazole is preferred unless resistance or non-albicans species are suspected, in which case voriconazole provides broader coverage.